



GENETIC DISORDERS — FULL MIND MAP

Biology Ch. 9.v · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Genetic vs Non-Genetic Disorders
2. Phenylketonuria (PKU) — Inborn Error of Metabolism
3. Bubble Baby Disease — SCID
4. Haemophilia — Sex-Linked Hereditary Disorder
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6. X-Linked Disorders & Muscular Dystrophy
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1. Genetic vs Non-Genetic Disorders

- Source ke according Night blindness genetic disease nahi hai; iska cause Vitamin A deficiency diya hai. Albinism, Haemophilia aur Colour blindness ko genetic diseases bataya gaya hai. [Q1]
- Another source comparison mein Down syndrome, Haemophilia aur Sickle Cell Anaemia hereditary/genetic disorders hain, jabki Irritable Bowel Syndrome (IBS) ko genetic disorder nahi bataya gaya hai. [Q10]

2. Phenylketonuria (PKU) — Inborn Error of Metabolism

- Phenylketonuria (PKU) ko inborn error of metabolism ka example diya hai. “Error” ka meaning inherited lack of an enzyme hai; source phenylalanine hydroxylase deficiency ko phenylalanine → tyrosine conversion se link karta hai. [Q2]
- Enzyme deficiency ki wajah se phenylalanine blood/tissue fluids mein high ho sakta hai; source progressive mental retardation / mental idiocy ko important phenotypic effect batata hai. [Q2, Q3]

3. Bubble Baby Disease — SCID

- “Bubble Baby Disease” = Severe Combined Immunodeficiency (SCID), ek rare genetic disorder. Source ke mutabik functional T-cells aur B-cells ka development disturbed hota hai aur affected baby ko fatal infections se bachane ke liye germ-free plastic bubble mein treat/keep kiya jaata hai. [Q4]

4. Haemophilia — Sex-Linked Hereditary Disorder

- Haemophilia source mein genetic sex-linked hereditary defect hai. Source wording: women carry it while it appears in men; affected person mein blood clotting proper nahi hoti aur clotting mein bahut zyada time lag sakta hai. [Q5, Q6, Q8, Q9]
- Royal haemophilia ko sex-linked genetic disease bataya gaya hai. Source Queen Victoria association ke kaaran ise “royal haemophilia / royal disease” bhi kehta hai. [Q5, Q7]

5. Down Syndrome — Trisomy 21

- Down Syndrome = Trisomy 21. Source ke according chromosome 21 ki third copy ki presence se disorder hota hai; chromosome-number change ke kaaran total chromosome count 47 ho jaata hai. [Q11, Q12, Q13, Q14]
- Source Down syndrome ko physical growth delays, mild-to-moderate intellectual disability aur characteristic facial features se associate karta hai; explanation mein historical term “Mongolian idiocy” bhi diya gaya hai. [Q11, Q14]

6. X-Linked Disorders & Muscular Dystrophy

- Source Q15 mein sex chromosome par recessive gene ki expression se caused disorder ka answer Muscular dystrophy diya hai; explanation ise X-chromosome-linked hereditary disease aur progressive muscle weakness/loss se link karti hai. [Q15]

7. Disease-Type Matching

- Correct matching: Haemophilia → Genetic disease; Diabetes → Hormonal disorder (insulin deficiency); Rickets → Deficiency disease (Vitamin D deficiency); Ringworm → Fungal infection. [Q16]
- Correct hereditary set source mein Colour blindness + Haemophilia + Sickle cell anaemia hai. Diphtheria/Pneumonia/Leprosy bacterial; AIDS viral; Syphilis/Gonorrhoea bacterial; Polio/Japanese B encephalitis viral aur Plague bacterial bataya gaya hai. [Q17]
- Vitamin–disease matching trap: Vitamin A deficiency → Night blindness, Colour blindness nahi. Thiamine (B1) → Beriberi; Ascorbic acid (C) → Scurvy; Vitamin K → blood clotting. [Q19]

8. Colour Blindness — X-Linked Inheritance

- Colour blindness source mein X-linked recessive trait hai. Men mein sirf one X-chromosome hone ki wajah se X-linked recessive disorders females ke comparison mein more frequent hote hain; fathers X-linked trait sons ko directly pass nahi karte. [Q18]
- Colour-blind man (XcY) + completely normal woman (XX): source explanation ke according immediate sons normal aur daughters carrier-normal hoti hain; immediate children mein colour blindness probability 0% hai. [Q20, Q21]
- Source Q20 ka marked answer “sons of daughters” hai: carrier daughters aage normal male se marry karen to unke kuch sons colour blind ho sakte hain. Isliye question ka focus next generation/grandsons par hai. [Q20]
- Colour-blind woman (XcXc) + normal husband (XY): source ke according son colour blind hoga, jabki daughter affected X inherit karke carrier hogi par colour blind nahi hogi. [Q22]

EXAM TRAP — Colour-Blindness Generations

- ▶ Q20 immediate children ke baare mein nahi, “symptoms generated in sons of daughters” ko key karta hai; Q21 same parental cross mein immediate children ka actual colour-blind probability 0% deta hai. Dono source explanations ko generation-wise padhna zaroori hai. [Q20, Q21]

9. Thalassemia — Haemoglobin & Blood Disorder

- Thalassemia inherited/genetic blood disorder hai. Source ke according body abnormal form ya inadequate amount of haemoglobin banati hai; RBC destruction severe anaemia tak lead kar sakta hai. [Q23]
- Isliye Thalassemia mein affected system/tissue context ka answer Blood hai, aur synthesis-related answer Haemoglobin hai. [Q23, Q24]